Task Prioritization Framework for Kinesthetic Teaching of a Free-standing Humanoid Robot

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Abstract— **This paper presents a framework for whole-body motion control of a free-standing humanoid robot, for kinesthetic teaching through pHRI (Physical Human Robot Interaction). The proposed method is based on a task-oriented control through task prioritization which enables execution of multiple robot tasks in parallel, exploiting system redundancy. The tasks with lower priority are handled without affecting execution of the primary ones. The proposed control algorithm provides compliant behavior of the robot end-effectors, in order to enable their physical guiding by demonstrator for skill acquisition, while preserving dynamic balance of a free standing humanoid, by keeping the zero-moment point (ZMP) inside the support area, in the presence of multiple constraints and contacts. The framework takes into account torque saturation, joints' mechanical limits and exploits system redundancy to maintain the desired realization of imposed tasks. In the proposed approach all imposed tasks and constraints, including the maintenance of dynamic balance, are handled without direct influence of a human, which allows a user to focus on the end-effector guidance. The proposed method has been tested by a simulation of manipulation task on a humanoid robot. Hip-ankle strategy, for the maintenance of dynamic balance, emerges as a consequence, without explicit programing.**

I. INTRODUCTION

In the past decades robots were exclusively used in industry applications, where they performed, in advance programmed, tasks in highly structured environment. Nowadays, the fields of robot applications broaden, from service and rescue robots to agriculture, medicine and entertainment. The current trend in the robotics research community is to achieve full living and working coexistence of humans and robots. Since robots should operate in an environment which is adapted to the human needs and have to use tools which are ergonomically designed for humans, it is convenient to choose an anthropomorphic robots. Humanoid robots, with such structure, are highly redundant dynamic systems with many DOFs (Degrees Of Freedom) and it is extremely demanding to program them to perform a complex task in an unstructured environment. Having in mind that humanoids perform biped locomotion, execution of the most tasks and skills assumes dynamical balance preservation. Programming such simultaneous execution of different tasks requires an expert programmer, with a strong background in robotics.

Thus, a new way of programming robots, which would enable every-day users without technical background to use and program robots has been imposed as a fundamental requirement. This is particularly the case if some new tasks and skills should be acquired.

Learning from demonstration (LfD) is a valuable method for teaching a robot complex tasks and novel skills without tedious and demanding manual programming. Good survey on this topic can be found in [1, 2]. There are several methods to perform skill demonstration. One popular way is to use a motion capture system to record the motion of a human demonstrator and then to retarget (map) that motion to the robot's kinematic structure. This approach provides the ability to teach the robot complex tasks which involve many DOFs. The problem of retargeting recorded motion is not trivial. One solution, called Direct Marker Control, was proposed in [3]. The selected set of control points on the robot is connected virtually with translational springs to appropriate measured marker points. In this way every movement of the demonstrator has been mapped to the robot. Beside the high cost, there are several other drawbacks of approach requiring motion capture system. It needs a fixed laboratory setup for motion recording, so it is necessary to transport the robot to the lab to teach it a new skill. Also it is very hard to demonstrate and retarget motion for a robot whose kinematic structure is significantly different from a human's.

Another approach [4, 5], records basic, reference motion from a motion capture system and then uses pHRI (Physical Human Robot Interaction) to refine that motion. Impedance control provides a desired compliant behavior for interaction.

An alternative approach to motion capturing is kinesthetic teaching. In such a case, skill demonstration is performed through physical interaction of operator (user) and robot. However, it can be very demanding for a demonstrator to physically guide the gripper of a highly redundant robot and to control the configuration of all its joints. Some early works on kinesthetic teaching handled this problem by either deactivating the controlled joints or setting very low servo gains [6], [7]. As a consequence, these approaches often lead to unsynchronized and unnatural motions because the teacher moves the motors one by one rather than demonstrating natural coordinated movements [8]. In [9, 10] combination of kinesthetic teaching with supervised learning was proposed. A neural network was used to learn the inverse kinematics for different working areas. In the programming phase assisted gravity compensation control was used, allowing a user to guide the robot's end-effector while using the given redundancy resolution encoded in a neural network.

In traditional industrial applications where robots are fixed to the ground, manipulation and interaction prob-

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lems could be treated separately from the task of preserving dynamic balance [11]. However, considering that physical influence of an operator on a humanoid could endanger robot's dynamic balance, it is necessary to treat both of these problems simultaneously. The approach proposed in [12] treats simultaneous execution of different tasks through a hierarchical prioritization framework. The same authors proposed later in [13] a unified framework for whole body humanoid robot control with multiple constraints and contacts. This idea seems promising in the context of kinesthetic teaching. An approach, which uses a multi-priority kinematic controller (so called Task Transition Control), is proposed in [14]. With this approach, the user has the ability to do physical refinement of both end-effector and null-space motion.

Only few papers addressed the problem of kinesthetic teaching of a full-body free standing humanoid robot. In [15] an integrated approach for upper-body kinesthetic teaching of a free-standing humanoid was proposed. The proposed method treats the balancing problem and upper-body control with different control strategies. Balance preservation was provided using Reaction Null Space method while the upper-body was guided via kinesthetic teaching. The multi-priority controller approach, proposed in [14] was extended towards kinesthetic teaching of a bipedal humanoid robot [16]. External forces, detected by a momentum based disturbance observer, were integrated in the bipedal balancing algorithm and were used to parameterize the interaction controller.

In this paper a novel approach for assisted kinesthetic teaching of a free-standing humanoid robot, through pHRI has been proposed. It is based on the task-oriented control and uses the task prioritization framework [17], which exploits system redundancy, to perform multiple equality and inequality-type tasks with different priorities in parallel. The framework presented in [17] is an extension and improvement of the work given in [12], extended for nonholonomic tasks and systems. This paper is an extension of our previous work [18], in which the framework was used for kinesthetic teaching of robot upper-body. To the best of our knowledge, the framework presented in this paper is the first kinesthetic teaching approach for full-body free-standing humanoid robot that treats all the tasks and constraints, including both balance preservation and physical guiding of end-effector, in a unified manner.

Applied control methodology is based on successive null-space projections where each task is performed in the null-space of all higher priority tasks. The framework is able to handle both holonomic and nonholonomic constraints equally, as well as the unilateral constraints. This control system provides assistance during kinesthetic teaching and allows the demonstrator to focus only on the guidance of the robot end-effector. In the presented simulation case study the robot was standing and its end-effectors were independently guided to reach two boxes of unknown weight. Under the assumption of having a force/torque sensor, the payload was estimated and then the end-effectors were guided in such a manner to transfer the boxes to another place. The Hip-Ankle strategy for balance preservation emerged as a consequence of required tasks without being explicitly programmed.

The rest of the paper is organized as follows. In section II a brief theoretical background on the task prioritization framework is given. In section III the robot model, used in simulation, is briefly described, while in section IV the results from the simulated case study are presented.

II. Introduction to Prioritization Framework

In this section just a short background for the prioritization framework is presented, while more details can be found in [17]. A robotic system is supposed to have n links. Coordinate, velocity, and acceleration at the i-th joint are denoted by $q_i, \dot{q}_i, \ddot{q}_i$ while its driving torque is denoted as τ_i. The system dynamics is governed with the following differential equation:

$$\mathbf{H}\ddot{\mathbf{q}} + \mathbf{h}_0 = \boldsymbol{\tau} + \mathbf{J}^T \mathbf{F}_{ext}, \tag{1}$$

where $\mathbf{H}$ is a positive definite inertia matrix, $\mathbf{h}_0$ is the vector which includes the velocity and gravitational effects, $\ddot{\mathbf{q}}$ is the vector of joint accelerations, $\boldsymbol{\tau}=[\tau_1, \tau_2,..., \tau_n]^T$ is the vector of joint driving torques, $\mathbf{J}$ is the Jacobian matrix and $\mathbf{F}_{ext}$ is the vector of generalized forces acting on the robot from the environment. The problem is to control the system described by (1) to accomplish p tasks, with different priorities, in parallel. The proposed framework is able to handle both equality and inequality-type tasks which can be represented with one of the following expressions:

$$\mathbf{A}_i\ddot{\mathbf{q}} = \mathbf{b}_i \text{ or } \mathbf{A}_i\ddot{\mathbf{q}} \preceq \mathbf{b}_i . \tag{2}$$

The size of the matrix $\mathbf{A}_i$ is $m_i \times n$, with $m_i \leq n$, and the length of the vector $\mathbf{b}_i$ is m_i. This means that number of tasks must be less or equal to the number of DOFs. A lower priority task is projected in the null-space of all higher priority tasks, and it does not interfere with their execution. The form of task representation (2) can be applied to wide variety of tasks. Any task written in the form $\mathbf{f}(\mathbf{q},t) = \mathbf{x}_{des}$ or $\mathbf{f}(\mathbf{q},\dot{\mathbf{q}},t) = \mathbf{x}_{des}$ can be represented in the form (2) by differentiating, with respect to time, once (for the second case) or twice (for the first case). An example of the equality-type task is a reaching task. Inequality tasks of the form $\mathbf{f}(\mathbf{q},t) \preceq \mathbf{x}_{limit}$ and $\mathbf{f}(\mathbf{q},\dot{\mathbf{q}},t) \preceq \mathbf{x}_{limit}$ can be reduced to the form (2) using Taylor expansion up to the second derivative. The proposed framework is well-suited for both prioritization of equality, and inequality-type tasks, like avoiding slipping or keeping the ZMP within the support area. A system can be constrained and the constraints can be written in the form $\mathbf{A}_0\ddot{\mathbf{q}} = \mathbf{b}_0$ [19, 20]. The objective of the framework is to calculate an appropriate driving torque vector $\boldsymbol{\tau}$ for the system (1) which fulfills p tasks given by (2). The procedure for calculating driving torques, based on [17], is shown in Algorithm 1. The torques are calculated by successively solving the optimization problem for each task. The modified task matrices are $\mathbf{B}_i=\mathbf{A}_i\mathbf{H}^{-1/2}$. The modified vector of joint acceleration is $\ddot{\mathbf{r}} = \mathbf{H}^{1/2}\ddot{\mathbf{q}}$ while the modified velocity and gravity vector is $\mathbf{p}=\mathbf{H}^{-1/2}\mathbf{h}_0$. The null space of constraint is $\mathbf{N}_c$ while the modified vector of torques induced by the constraint is denoted by $\mathbf{T}_c$. The null space of all tasks preceding the current task is denoted by $\mathbf{N}_{prev}$ and the modified torque vector which needs to be

applied in order to fulfill all tasks preceding the current one is $\mathbf{T}_{prev}$. The matrix $\mathbf{C}$ and vector $\mathbf{d}$ are used in order to include inequality tasks in the framework. The set S contains the indexes of the inequalities that are strictly fulfilled after the optimization and the set $\overline{S}$ contains the indexes of the inequalities that have reached the equality sign.

Algorithm 1. Procedure for determining control torques

1: $\mathbf{N}_c \leftarrow \mathbf{I}; \mathbf{T}_c \leftarrow \mathbf{0}; \ \mathbf{C} \leftarrow [\ ]; \mathbf{d} \leftarrow [\ ];$

2: **if** system constrained **then** $\mathbf{N}_c \leftarrow \mathbf{I} - \mathbf{B}_0^+ \mathbf{B}; \mathbf{T}_c \leftarrow \mathbf{B}_0^+ (\mathbf{b}_0 + \mathbf{B}_0 \mathbf{p});$

3: **for** Each task i **do**

4: **if** task of inequality type **then**

5: **if** system feasible **then**

6: $\mathbf{C} \leftarrow \begin{bmatrix} \mathbf{C} \\ \mathbf{B}_i \end{bmatrix}; \mathbf{d} \leftarrow \begin{bmatrix} \mathbf{d} \\ \mathbf{b}_i \end{bmatrix};$ **continue;**

Solve: minimize $\| \mathbf{B}_i \ddot{\mathbf{r}} - \mathbf{b}_i \|_2^2$

7: subject to: $\ddot{\mathbf{r}} + \mathbf{p} = \mathbf{T}_c + \mathbf{N}_c \mathbf{T}_{prev} + \mathbf{N}_c \mathbf{N}_{prev} \mathbf{u}$, $\mathbf{C}\ddot{\mathbf{r}} \preceq \mathbf{d}$

8: $\mathbf{T}_{prev} \leftarrow \mathbf{T}_{prev} + \mathbf{N}_{prev} \mathbf{u}; \overline{S} \leftarrow \{k \mid \mathbf{C}_k \ddot{\mathbf{r}} = d_k\}; S \leftarrow \{k \mid \mathbf{C}_k \ddot{\mathbf{r}} < d_k\};$

9: $\mathbf{N}_{prev} \leftarrow \mathbf{N}_{prev} \left(\mathbf{I} - \left(\begin{bmatrix} \mathbf{B}_i \\ \mathbf{C}_{\overline{S}} \end{bmatrix} \mathbf{N}_c \mathbf{N}_{prev} \right)^+ \begin{bmatrix} \mathbf{B}_i \\ \mathbf{C}_{\overline{S}} \end{bmatrix} \mathbf{N}_c \mathbf{N}_{prev} \right); \mathbf{C} \leftarrow \mathbf{C}_S; \mathbf{d} \leftarrow \mathbf{d}_S;$

10: **Endfor**

11: $\boldsymbol{\tau} \leftarrow \mathbf{H}^{1/2} \mathbf{T}_{prev}$

III. Simulation Model

The robot is modeled as a set of kinematic chains connected to the base link, the trunk segment. All links are interconnected with simple 1 DOF rotational joints. The joints with more than one DOF (shoulder and wrist) are modeled as a series of one DOF rotational joints connected by massless and lengthless links. The robot's kinematic structure is shown in Fig. 1.

The presented kinematic model consists of five kinematic chains. Two chains comprise the right and left leg while the third and fourth chains comprise the right and left arm. The last, fifth chain represents the head. The position and orientation of the base link in space is defined with three translational and three angular coordinates $\mathbf{X} = [x, y, z, \varphi, \theta, \psi]^T$.

The state vector $\mathbf{Q}$ consists of the position and orientation of the base link and the n elements vector $\mathbf{q}$ consisting of the relative angles between two adjacent links:

$$\mathbf{Q} = \left[\mathbf{X}^T \mathbf{q}^T \right]^T = [x, y, z, \varphi, \theta, \psi, q_1, q_2, ..., q_n]^T, \quad (3)$$

The robot legs have 7 DOFs each while the arms have 6 DOFs. The neck has 1 DOF and there are additional 6 DOFs for position and orientation of the base segment. The overall number of DOFs is therefore 33. Since the base link is unactuated, the vector of joint driving torques at the joints is:

$$\boldsymbol{\tau} = [0, 0, 0, 0, 0, 0, \tau_1, \tau_2, ..., \tau_n]. \quad (4)$$

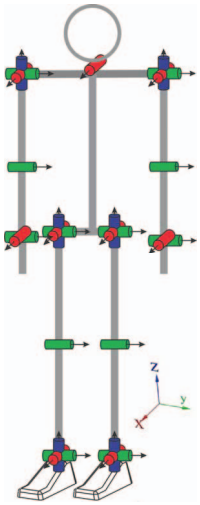


Figure 1. Kinematic structure of robot upper-body

The robot foot is modeled as a rigid body with a viscoelastic layer, using Kelvin-Voigt model. This kind of contact model is able to model sliding, foot rotation and deformation of the viscoelastic layer. The contact between the foot and ground is modeled as a matrix of single point contacts [21].

IV. Simulation Results

The performance of the proposed approach has been tested with simulation. The proposed prioritization framework was applied on a 27 joints free-standing humanoid robot in the kinesthetic teaching process. In this section a brief description of the simulation case study is given with the formal mathematical description of all the tasks and constraints imposed on the robot. At the end of the section the simulation results are presented. Simulated motion consists of five phases. At the initial moment, the robot stands still with both feet on the ground. Joint torques are constant, and correspond to the gravitational load of the joints. Two boxes weighing 5 kg are on the table, but the weight is unknown to the robot. In the first phase both of the robot's end-effectors are physically guided by operator (whose influence is represented by appropriate external forces applied on the grippers) to reach the two boxes on the table in front of the robot. Both external forces are realized with a spring-damper model between the current position of the end-effector and the target position in space, with force saturation at ±10N. The forces, applied to the left and right hand ($\mathbf{F}_{extl}$, $\mathbf{F}_{extr}$), are described as:

$$\mathbf{F}_e = \mathbf{K}_p (\mathbf{x}_d - \mathbf{x}) - \mathbf{K}_d \mathbf{J}\dot{\mathbf{q}}$$
$$\mathbf{F}_{ext} = \frac{\min(\|\mathbf{F}_e\|, 10)}{\|\mathbf{F}_e\|} \mathbf{F}_e, \quad (5)$$

where $\mathbf{K}_p$ and $\mathbf{K}_d$ are positive definite, diagonal gain matrices, $\mathbf{x}_d$ is a desired position in Cartesian space, while $\mathbf{x}$ is current end-effector position. The grasping itself was not addressed in the paper and was simulated as

a rigid contact between the robot end-effector and the box. In the second phase the robot manually lifts the boxes and holds them still to estimate their weight in order to differentiate it from the guiding external force of the human demonstrator. It was assumed that the robot has a force/torque sensor in the wrist. The measured external force is $\mathbf{F}_{\text{ext}} = \text{m}\mathbf{g}$, so the mass can be estimated easily. In the next two phases the robot hands, holding the boxes, are physically guided to lift the boxes to the raised platform on the table. In the last phase the robot brings the first box down on the platform and then places the second box on top of the first one. The contact between two boxes and between the first box and the platform was simulated the same way as the contact between the foot and the ground [22]. At the end the robot returns to its initial pose. In all phases, except in the second one, the desired impedance is provided on the robot end-effectors and the robot moves only under the influence of external forces or the weight of the payload.

All the tasks, to be performed, can be divided in two groups. The first group comprises the tasks that ensure structural integrity of the robot (to keep all joints torques within predefined ranges, to keep joint coordinates within their mechanical limits, to avoid self-collision and robot overturn, …) while to the second group belong the tasks which should provide desired robot behavior. All tasks of the first group are assigned to the highest rank. The highest priority task is to keep the driving torques at all joints between their saturation limits. This is an inequality type task which can be written as:

$$\begin{bmatrix} \mathbf{H}^{1/2} \\ -\mathbf{H}^{1/2} \end{bmatrix} \mathbf{T}_{p+1|prev} \le \begin{bmatrix} \tau_{sat} \\ \tau_{sat} \end{bmatrix}, \tag{6}$$

where $\mathbf{T}_{p+1|prev}$ is the torque which fulfills all the tasks from 1 to p. The second most important task is to keep all joint coordinates between their mechanical limits which can also be written as an inequality:

$$\frac{\mathbf{T}^2}{2} \begin{bmatrix} 0_{27x6} & \mathbf{I}_{27x27} \\ 0_{27x6} & \mathbf{I}_{27x27} \end{bmatrix} \ddot{\mathbf{Q}} \le \begin{bmatrix} \mathbf{q}_u - \mathbf{q} - \dot{\mathbf{q}}T \\ -\mathbf{q}_l + \mathbf{q} + \dot{\mathbf{q}}T \end{bmatrix}, \tag{7}$$

where $\mathbf{q}_u$ and $\mathbf{q}_l$ are the upper and lower limits that joint coordinates should not exceed. $\mathbf{T}$ represents the preview time interval. Zero submatrices (0_{27x6}) in (7) correspond to the first 6 DOFs, position and orientation of the fixed base segment. The last task, from the first group, is to maintain the foot ground contact firm. This means that contact should be such to prevent rotation or sliding of the foot over the ground and to prevent separation of the foot from the ground. For a point contact, with Columb friction, the contact force must lie inside a friction cone. Condition that the contact force is inside friction cone (approximated by n-sided friction pyramid) can be written as:

$$\mathbf{S}_{\mu s \times 3} \mathbf{F}_{3\times 1} \succ \mathbf{0}_{s\times 1}, \tag{8}$$

where $\mathbf{S}_{\mu s \times 3}$ is the matrix that defines the friction pyramid and F is the force vector [22]. For rectangular robot feet, the surface contact with the ground can be considered as contact between four points at the corners of the foot and the ground. The total wrench calculated for the middle of the foot M is:

$$\begin{bmatrix} \mathbf{F}_M \\ \mathbf{M}_M \end{bmatrix} = \begin{bmatrix} \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \mathbf{F}_4 \\ \mathbf{r}_1^M \times \mathbf{F}_1 + \mathbf{r}_2^M \times \mathbf{F}_2 + \mathbf{r}_3^M \times \mathbf{F}_3 + \mathbf{r}_4^M \times \mathbf{F}_4 \end{bmatrix}, \quad = \bar{\mathbf{G}}_M \bar{\mathbf{F}}_M \tag{9}$$

where $\bar{\mathbf{F}}_{\text{M}} = \left[\mathbf{F}_1^{\text{T}}, \mathbf{F}_2^{\text{T}}, \mathbf{F}_3^{\text{T}}, \mathbf{F}_4^{\text{T}}\right]^{\text{T}}$ is the vector of all four contact forces acting on the foot and $\bar{\mathbf{G}}_{\text{M}}$ is the contact matrix. Each of the forces needs to be inside its respective friction pyramid, given by (8). Combining eq. (9) with the friction pyramid constraint leads to:

$$\bar{\mathbf{S}}\bar{\mathbf{G}}_{\text{M}}^{\dagger} \begin{bmatrix} \mathbf{F}_{\text{M}} \\ \mathbf{M}_{\text{M}} \end{bmatrix} = \mathbf{Z} \begin{bmatrix} \mathbf{F}_{\text{M}} \\ \mathbf{M}_{\text{M}} \end{bmatrix} \succ 0 . \tag{10}$$

where $\bar{\mathbf{G}}_{\text{M}}^{\dagger}$ is the pseudo-inverse of the matrix $\bar{\mathbf{G}}_{\text{M}}$. $\bar{\mathbf{S}}$ is the matrix with submatrices $\mathbf{S}_{\mu s \times 3}$ on its diagonal and zero submatrices for other elements. The new matrix $\mathbf{Z}$ has been introduced and named dynamic balance matrix. If the previous condition is fulfilled then the foot-ground contact is proper. This means that foot-ground contact is over surface and that there is no sliding. Thus, the dynamic balance matrix is the generalization of ZMP concept, with included sliding. In the case of foot sliding the inequality (10) would be violated.

The tasks from the second group provide desired robot behavior. The first task of this group is to maintain the CoM above the desired position, which is in this example the initial position (x=0, y=0). This is an equality type task, which ensures robot static balance and whose task matrices can be written as:

$$\mathbf{A} = \mathbf{J}_{CM|xy}, \mathbf{b} = \ddot{\mathbf{x}}_{CM|des} - \dot{\mathbf{J}}_{CM|xy}\dot{\mathbf{Q}}, \tag{11}$$

where $\mathbf{J}_{\text{CM|xy}}$ represents the Jacobian associated with the CoM in the x and y directions, while $\ddot{\mathbf{x}}_{CM|des}$ denotes the desired acceleration of CoM. For physical kinesthetic teaching it is necessary to provide compliant behavior of the robot end-effectors. The second task of this group is to provide desired Cartesian impedance on the end-effectors. The impedance control was used to achieve controlled gravity compensation like in [10]. The desired impedance model is given with:

$$\mathbf{H}_m\left(\ddot{\mathbf{x}} - \ddot{\mathbf{x}}_d\right) + \mathbf{D}_m\left(\dot{\mathbf{x}} - \dot{\mathbf{x}}_d\right) + \mathbf{K}_m\left(\mathbf{x} - \mathbf{x}_d\right) = \mathbf{F}_{ext}, \tag{12}$$

where $\mathbf{H}_m$ is the desired inertia matrix, $\mathbf{D}_m$ is the desired damping matrix and $\mathbf{K}_m$ is the desired stiffness matrix. In each time instant desired position ($\mathbf{x}_d$) is chosen to be the same as the current one, while the desired Cartesian velocity and acceleration ($\dot{\mathbf{x}}_d$, $\ddot{\mathbf{x}}_d$) are chosen to have zero value. This way the robot tries to keep its

current position, like in a gravity compensation mode, but provides enough compliance to be guided by an external force. Considering this, and taking the direct kinematic relation $\dot{\mathbf{x}} = \mathbf{J}\dot{\mathbf{Q}}$ into account, the desired acceleration can be obtained from (12):

$$\ddot{\mathbf{x}}_{des} = \begin{bmatrix} \mathbf{H}_m^{-1}\left(-\mathbf{D}_m\mathbf{J}_{left}\dot{\mathbf{Q}} + \mathbf{F}_{extl}\right) \\ \mathbf{H}_m^{-1}\left(-\mathbf{D}_m\mathbf{J}_{right}\dot{\mathbf{Q}} + \mathbf{F}_{extr}\right) \end{bmatrix}, \qquad (13)$$

where $\mathbf{F}_{extl}$ and $\mathbf{F}_{extr}$ are external forces applied to the left and the right hand, respectively and where $\mathbf{J}_{left}$ and $\mathbf{J}_{right}$ represent the Jacobian matrices associated with linear movement of the left and the right hand of the robot. The damping term stops the end-effector drift after external force decreases to zero. The task (providing desired impedance) can be represented in the following form:

$$\mathbf{A} = \begin{bmatrix} \mathbf{J}_{left} \\ \mathbf{J}_{right} \end{bmatrix}, \ \mathbf{b} = \ddot{\mathbf{x}}_{des} - \begin{bmatrix} \dot{\mathbf{J}}_{left} \\ \dot{\mathbf{J}}_{right} \end{bmatrix}\dot{\mathbf{Q}}, \qquad (14)$$

The task with the lowest priority implemented in the presented simulation was to achieve the desired configuration. For this task a proportional and derivative (PD) controller was employed to calculate the desired Cartesian acceleration:

$$\ddot{\mathbf{x}}_{des} = k_p\left(\mathbf{q}_0 - \mathbf{q}\right) - k_d\dot{\mathbf{q}}, \qquad (15)$$

where k_p and k_d are proportional and derivative gains for the considered task. The task matrices can be written as:

$$\mathbf{A} = \begin{bmatrix} 0_{27x6} & \mathbf{I}_{27x27} \end{bmatrix}, \ \mathbf{b} = \ddot{\mathbf{x}}_{des}. \qquad (16)$$

The elements of vector $\mathbf{q}_0$ (desired robot configuration) are positions of joints at the initial moment.

Simulation results are shown in Figs. 2-4. The stick diagram of the robot, at the end of each phase, is shown in Fig. 2. It can be noted that the hip-ankle strategy emerged when the robot was reaching the boxes. This strategy was not explicitly programmed, but emerged as a consequence of all the tasks defined in the framework. In the example presented only end-effectors were externally guided, while the proposed framework handled all the constraints imposed and ensured task realization while preserving the dynamical balance of the robot.

In Fig. 3 position of the ZMP and CoM, during the complete movement, is shown. The red rectangle represents the left footprint and the blue the right one. The position of the ZMP is denoted with red crosses. It can be seen that the ZMP stays inside the support area during the whole motion. Its position changes mostly along the x-axis, because the robot is leaning forwards to reach the boxes. One of the tasks in the framework was to keep the CoM in its initial position. It can be seen from Fig. 3 that this task is fulfilled. In Fig. 4 the position of the left end-effector in x-, y- and z- direction is given.

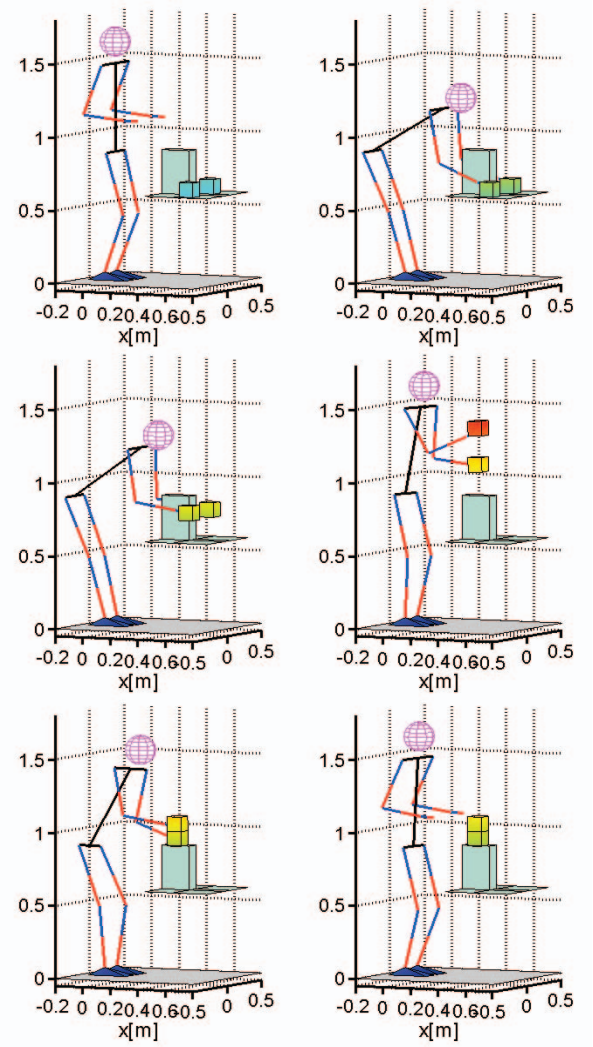


Figure 2. Stick diagram of the robot at the end of each phase of motion

The dashed line in Fig. 4 represents the referent position for each phase of motion, while the full line shows the end-effector trajectory. It can be seen from Fig. 4 that the robot end-effector converges to its referent position in each phase. The end-effector moves under the influence of the external force, which represents the physical interaction with human demonstrator. The robot reaches the desired position without overshoot and eliminates the steady-state error.

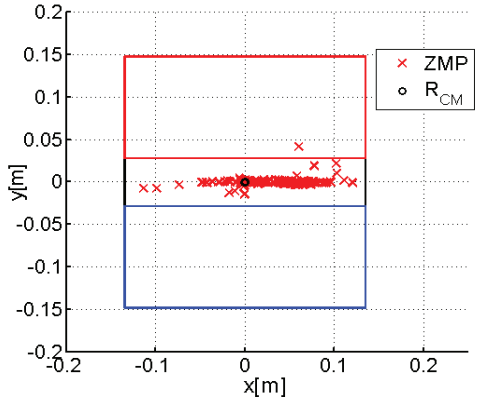


Figure 3. Position of ZMP (red crosses) and CoM (black circle) during motion. Support area is shown with a black rectangle

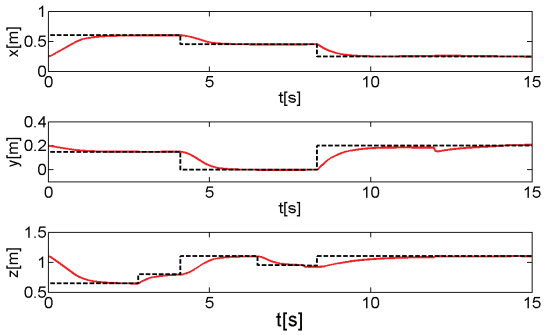


Figure 4. Trajectory of the left end-effector in x-, y- and z- direction (red line) with referent positions for each phase (dashed line)

V. CONCLUSION

In this paper a novel control method for a skill demonstration through the physical interaction for full-body free-standing humanoid robots was presented. The proposed method is based on the task prioritization framework which exploits system redundancy to handle multiple tasks in parallel. Each task is performed in the null-space of all the tasks with higher priorities, so it does not interfere with their execution. The proposed framework ensures compliant behavior of the robot end-effectors and enables the demonstrator to focus on guiding the end-effector to teach the robot a new skill, while the framework preserves dynamic balance and provides a desired configuration of all joints, taking into account torque saturations and joint limits. The framework has been tested in simulation on a free-standing humanoid robot. The results have shown that the proposed framework is able to provide a desired impedance of end-effectors, in presence of constraints and multiple contacts, while preserving robot dynamic balance. In the future work the verification of the proposed framework on the experimental platform will be addressed as well as the parallelization of the optimization process in order to reduce the computational cost.

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REFERENCES

[1] A. Billard, S. Calinon, R. Dillmann and S. Schaal, "Robot Programming by Demonstration," in *Springer Handbook of Robotics*, Berlin, Germany: Springer Berlin Heidelberg, 2008, ch. 59, pp. 1371-1394.

[2] B. Argall, S. Chernova, M. Veloso and B. Browning, "A Survey of Robot Learning from Demonstration," *Robotics and Autonomous Systems*, vol. 67, pp. 469-483, 2009.

[3] C. Ott, D. Lee and Y. Nakamura, "Motion Capture based Human Motion Recognition and Imitation by Direct Marker Control," Proc. 8th IEEE-RAS International Conference on Humanoid Robots, Daejeon, Korea, pp. 399-405, Dec 1-3, 2008.

[4] D. Lee, C. Ott and Y. Nakamura, "Mimetic Communication Model with Compliant Physical Contact in Human–Humanoid Interaction," *The International Journal of Robotics Research*, vol. 29, no. 13, pp. 1684-1704, 2010.

[5] D. Lee and C. Ott, "Incremental Motion Primitive Learning by Physical Coaching Using Impedance Control," Proc. 2010 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Taipei, Taiwan, pp. 4133-4140, Oct 18-22, 2010.

[6] S. Calinon and A. Billard, "Active teaching in robot programming by demonstration," in IEEE International Conference on Robot and Human Interactive Communication, Jeju, Korea, pp. 702–707, Aug 26-29, 2007.

[7] T. Inamura, N. Kojo and M. Inaba, " Situation Recognition and Behavior Induction based on Geometric Symbol Representation of Multimodal Sensorimotor Patterns," Proc. 2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Beijing, China, pp. 5147-5152, Oct 9-15, 2006.

[8] D. Lee, C. Ott, Y. Nakamura and G. Hirzinger, "Physical Human Robot Interaction in Imitation Learning," Proc. 2011 IEEE International Conference on Robotics and Automation, Shanghai, China, pp. 3439-3440, May 9-13, 2011.

[9] S. Wrede, C. Emmerich, R. Grünberg, A Nordmann., A. Swadzba and J. Steil, "A User Study on Kinesthetic Teaching of Redundant Robots in Task and Configuration Space," *Journal of Human-Robot Interaction*, vol. 2, no. 1, pp. 56-81, 2013.

[10] J. J. Steil, C. Emmerich, A. Swadzba, R. Grünberg, A. Nordmann, S. Wrede, "Kinesthetic Teaching Using Assisted Gravity Compensation for Model-Free Trajectory Generation in Confined Spaces," in *Gearing Up and Accelerating Cross-fertilization between Academic and Industrial Robotics Research in Europe: Technology Transfer Experiments from the ECHORD Project*, Switzerland, Springer International Publishing, 2014, pp. 107-127.

[11] F. Nori, J. Peters, V. Padois, J. Babic, M. Mistry and S. Ivaldi, " Whole-body Motion in Humans and Humanoids," in PRoc. of the Workshop on New Research Frontiers for Intelligent Autonomous Systems (NRF-IAS), Venice, Italy, pp. 81-92, July 18-19, 2014.

[12] L. Sentis and O. Khatib, "Task-oriented control of humanoid robots through prioritization," in Proc. IEEE-Robotics and Autonomous Systems/RSJ International Conf. Humanoid Robots, Santa Monica, CA, USA, November 2004.

[13] O. Khatib, L. Sentis and J. Park, "A Unifed Framework for Whole-body Humanoid Robot Control with Multiple Constraints and Contacts," in Proc. European Robotics Symp, Prague, Czech Republic (Springer, Berlin, Heidelberg, Germany), pp. 303-312, July 2008.

[14] M. Saveriano, Sang-ik An and D. Lee, "Incremental Kinesthetic Teaching of End-Effector and Null-Space Motion Primitives," Proc. EEE International Conference on Robotics and Automation, 2015, (accepted)

[15] P. Kormushev, D. N. Nenchev, S. Calinon, D. G. Caldwell, "Upper-body Kinesthetic Teaching of a Free-standing Humanoid Robot," Proc. 2011 IEEE International Conference on Robotics and Automation, Shanghai, China, pp. 3970-3975, May 9-13, 2011.

[16] C. Ott, B. Henze, D. Lee, "Kinesthetic teaching of humanoid motion based on whole-body compliance control with interaction-aware balancing ," Proc. 2013 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Tokyo, Japan, pp. 4615-4621, Nov. 3-7, 2013.

[17] M. Nikolić, B. Borovac, M. Raković and S. Savić, "A Further Generalization of Task-Oriented Control Through Tasks Prioritization," *Int. J. Human. Robot*, vol. 10, no. 3, pp. 1-29, 2013.

[18] S. Savić, M. Nikolić, B. Borovac, M. Raković, "Assisted Kinesthetic Teaching of Humanoid Robot Upper-Body using Task Prioritization Framwork," Proc. 2nd International Conference on Electrical, Electronic and Computing Engineering (IcETRAN 2015), Srebrno Jezero, Serbia, June 8-11, 2015, (accepted)

[19] Y. Fan, R. Kalaba, H. Natsuyama and F. Udwadia, "Reflections on the Gauss principle of least constraint," *J. Optim. Theory Appl*, vol. 127, no. 3, pp. 475-485, 2005.

[20] F. E. Udwadia and R. E. Kalaba, *Analytical Dynamics: A New Approach*, Cambridge, UK, Cambridge University Press ,1996.

[21] M. Nikolić, B. Borovac, M. Raković, „Simulation of foot-ground contact for bipedal robots," Proc. of 23rd International Conference on Robotics in Alpe-Adria-Danube Region (RAAD), 2014, Smolenice, Slovenia, pp. 1-6, Sept 3-5, 2014.

[22] B. A. Borovac, M. N. Nikolić, "Prioritized Control of Humanoid Robot in Presence of Multiple Contacts," Proc. of 5th International Congress of Serbian Society of Mechanics, Arandjelovac, Serbia, June 15-17, 2015, (accepted)